

NOT ALL CHANGES MATTER: DECISION-ALIGNED DELTA SELECTION FOR SPEECH RECOGNITION

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ABSTRACT

Recent studies improve speech recognition adaptation by incorporating pretrained-to-fine-tuned representation shifts with other features, but treat these changes as a whole. Whether adapted recognizers rely on all shift coordinates remains underexplored. We propose **Decision-Aligned Delta Selection (DADS)**, a task-aware framework that identifies decision-supporting representation shifts by aligning feature changes with the adapted recognizer's decision sensitivity. We find that adaptation gains are concentrated in a subset of fine-tuning-induced representation shifts in the encoder's native output basis, rather than uniformly distributed across coordinates. A released L2-ARCTIC recognizer maintains nearly full accuracy after discarding 25% of shifts (10.58% vs. 10.50% WER), and DADS preserves substantially more performance than magnitude selection when retaining only half of the shifts (11.71% vs. 16.64% WER). Further reversion analyses show that the adapted recognizer relies unevenly on the induced shifts.

Index Terms— speech recognition, self-supervised learning, model adaptation, representation analysis, feature selection

1. INTRODUCTION

Modern speech recognition systems increasingly rely on large-scale self-supervised speech representations, which provide strong initialization for adapting models to new speakers, accents, and domains [1, 2, 3, 4, 5, 6]. Through supervised fine-tuning, these models acquire task-specific capabilities from limited labeled speech and achieve substantial improvements under distribution shifts [7, 8, 9]. Adaptation reshapes the learned representation space by introducing substantial differences between pretrained and adapted representations, yet how such information is organized within the representation space remains poorly understood.

Recent studies have demonstrated that differences between pretrained and adapted models contain transferable information. Task arithmetic transfers model behaviors through parameter differences, while TIES-Merging and DARE selectively manipulate such differences for model combination [10, 11, 12]. Similar ideas have also been explored in automatic speech recognition through task vector algebra and selective model merging [13, 14]. In representation space, *Mind the Shift* exploits delta embeddings, defined as the difference between pretrained and fine-tuned speech representations, to enhance child ASR; accent steering further utilizes representation differences between speech groups to steer adaptation behavior [15, 16].

Although existing methods vary in how representation differences are constructed and utilized, they generally treat representation shifts as a unified adaptation signal without considering how individual changes affect the adapted recognizer. Our key insight

is that shift relevance changes across the representation-to-decision boundary: from feature movement in the representation space to decision sensitivity in the adapted recognizer. A large shift may have limited impact if the recognizer is insensitive to that direction, while a smaller shift can be critical when aligned with a decision-sensitive direction.

Two observations support this decision-aligned view. First, fine-tuning-induced shifts exhibit highly non-uniform utility across native output coordinates. Under a fixed retention budget, preserving selected shifts can maintain adaptation performance, while removing different groups of the same size leads to substantially different recognition outcomes. Second, shift magnitude alone cannot explain adaptation utility. Useful shifts emerge from the interaction between representation movement and the recognizer's decision sensitivity, rather than from either factor independently.

Based on these observations, we propose **Decision-Aligned Delta Selection (DADS)**, a decision-aligned pruning framework for fine-tuning-induced representation shifts. DADS first decomposes adaptation into native output coordinates and evaluates each coordinate by measuring how its change affects the adapted CTC decision. Specifically, DADS combines shift magnitude with CTC objective sensitivity to identify decision-supporting shifts, then selectively retains informative changes while removing redundant adaptation signals. The intervention keeps the adapted CTC head and decoding fixed, so retention and reversion directly test which changes are sufficient and necessary for the existing recognizer (Fig. 1).

We evaluate DADS on L2-ARCTIC across adapted ASR models [17] and reveal substantial redundancy in fine-tuning-induced shifts. DADS retains 98.9% of the full adaptation gain using only 50% of native output coordinates, achieving 11.71% WER compared with 16.64% for magnitude-based selection. Counterfactual reversion further shows that high-utility shift groups are more decision-relevant, causing substantially larger degradation when removed than low-utility groups.

Our main contributions are summarized as follows:

1. We identify that adaptation utility is concentrated in a decision-dependent subset of native-coordinate fine-tuning shifts, showing why shift magnitude alone is insufficient.
2. We align representation displacement with adapted CTC sensitivity to select decision-supporting native-coordinate shifts and remove redundant changes without retraining.
3. We use frozen-head retention and reversion to establish which shifts the adapted recognizer functionally relies on and to verify the value of decision-aligned selection.

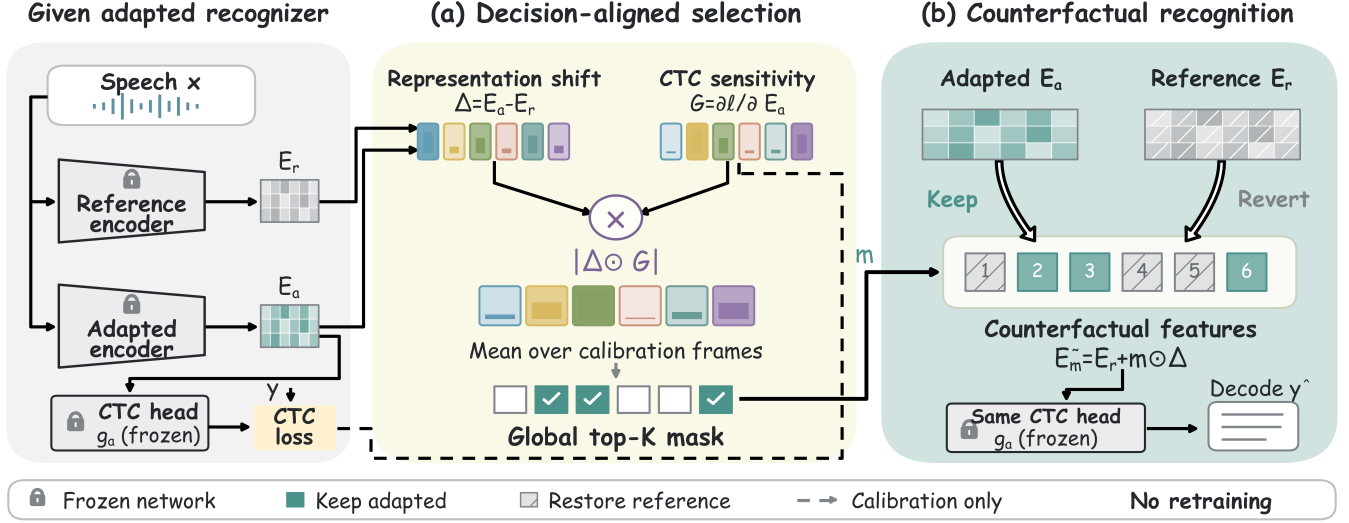


Fig. 1. Overview of DADS. (a) Shift Δ is combined with CTC sensitivity $G = \partial \ell / \partial E_a$ from the frozen adapted head. Averaging absolute elementwise products over valid training frames gives a global top- K native-coordinate mask. (b) Selected coordinates keep adapted values; others restore reference values. The same frozen head decodes without retraining. Labels and gradients are calibration-only; feature strips schematically show corresponding coordinates.

2. DECISION-ALIGNED DELTA SELECTION

2.1. Native-coordinate shift selection

DADS treats fine-tuning as a shift at the final encoder–CTC interface. It has two steps: (1) score each native coordinate by its movement and adapted CTC sensitivity; and (2) retain the highest-scoring changes while restoring the remainder to the reference representation. The adapted head is fixed after fine-tuning.

Learned native frame. We call the channel basis of the final encoder representation the native frame, i.e., the learned computational interface exposed to the adapted recognition head. Let $h = E(x)$ and $z = Wh + b$. For any orthogonal matrix Q , the equivalent parameterization

$$h' = Qh, \quad W' = WQ^T \quad (1)$$

preserves the input–output computation because $W'h' + b = Wh + b$, while a change is expressed as $\Delta h' = Q\Delta h$. The basis analysis in Section 4.4 tests whether DADS concentration follows the learned frame used by the trained recognizer. The frame is shaped jointly by the fine-tuned encoder and its readout; a learned input projection in a nonlinear decoder can absorb the same basis change before subsequent nonlinear operations.

2.2. Frozen-head intervention

As shown in Fig. 1, the reference and adapted encoders map utterance x to final-layer features $E_r(x), E_a(x) \in \mathbb{R}^{T_x \times d}$, where T_x is the number of output frames and d is the number of native coordinates. Their difference is

$$\Delta(x) = E_a(x) - E_r(x). \quad (2)$$

Both encoders receive the same waveform and have the same frame resolution. When the reference is the recorded initialization, Δ is the change induced by fine-tuning. We also apply the operation to a

released recognizer and a separately obtained reference, as specified in Section 3.

A binary mask $m \in \{0, 1\}^d$ retains K native coordinates of the change:

$$\tilde{E}_m(x) = E_r(x) + m \odot \Delta(x). \quad (3)$$

The mask is broadcast over frames and shared across utterances. Keeping every change gives the complete adapted representation (*Full*); keeping none gives the reference representation (*NoShift*).

Every condition uses the adapted affine CTC head $g_a(h) = W_a h + b_a$ and greedy decoding. NoShift therefore sends reference features through this same head. This comparison asks what the *adapted recognizer* needs from its feature changes. Because neither the encoder nor the head is retrained after masking, a WER difference directly records the effect of substituting the selected features. For each model, all masks are defined in its learned native final-layer frame exposed to W_a ; this frame is kept fixed across budgets and interventions.

The effect of a feature change therefore depends on how the recognition head uses its coordinate, motivating a score based on the recognition loss rather than shift magnitude alone.

2.3. Decision-aligned scoring

On a labeled subset $\mathcal{D}_u$ of the adaptation training data, we compute the loss at the complete adapted features:

$$\ell_{x,y} = \frac{1}{|y|} \mathcal{L}_{\text{CTC}}(g_a(E_a(x)), y), \quad (4)$$

where $|y|$ is the transcript length in characters. CTC sums over valid frame-to-label alignments [18]; no forced alignment is required for this score. The length normalization gives each utterance a loss per target character.

Undoing a change in native coordinate i perturbs each frame by $-\Delta_{t,i}$. In a first-order approximation, its effect on the loss involves

the product of the change and the corresponding gradient. Following Taylor saliency [19], DADS aggregates the absolute products:

$$U_i = \frac{1}{N_u} \sum_{(x,y) \in \mathcal{D}_u} \sum_{t=1}^{T_x} \left| \Delta_{t,i}(x) \frac{\partial \ell_{x,y}}{\partial E_{a,t,i}} \right|, \quad (5)$$

where $N_u = \sum_{(x,y) \in \mathcal{D}_u} T_x$. We take the absolute value at each frame before averaging, so positive and negative products do not cancel. Only valid frames contribute. The result is one score per native output coordinate, computed from a single pass through cached features and the fixed head.

DADS keeps the K native coordinates with the largest U_i . The score is high when a coordinate both moves during adaptation and affects the adapted recognition loss.

For comparison, *Magnitude* replaces the product in Eq. (5) with $|\Delta_{t,i}|$, while *Gradient* uses $|\partial \ell_{x,y} / \partial E_{a,t,i}|$. Both use the same valid-frame average and retain the same number of coordinates. *Random* selects exactly K coordinates uniformly. We also undo the highest- or lowest-scoring quarter to test the necessity of the ranking.

3. EXPERIMENTAL SETUP

3.1. Datasets

We evaluate DADS on L2-ARCTIC [17] using the speaker-disjoint splits of Thai et al. [7]. Each split contains 18 speakers for training and development and six held-out speakers for testing, covering six first-language groups. Folds 0–2 contain 675, 686, and 691 test utterances, respectively. Rankings are computed from training speech and evaluated on unseen speakers.

3.2. Adapted models

We study four adapted recognizers. W2V2-F0 is the released Fold 0 wav2vec 2.0 Large recognizer trained with CTC and supervised contrastive learning [20, 7]; its reference is the LibriSpeech-960h wav2vec 2.0 Large checkpoint [21]. W2V2-F1/F2 use the same backbone and objectives on Folds 1/2, while D2V-F0 adapts data2vec Audio Large on Fold 0 with CTC. The latter three use their recorded initializations as references. All models use 16-kHz input, 1024-dimensional final-layer features, and 32-symbol CTC heads.

We warm up the head for one epoch at 3×10^{-6} , then jointly adapt with frozen convolutional features at 10^{-5} . The best development-WER checkpoint is selected within 40 epochs. W2V2-F1/F2 use two GPUs with batch size 24 per GPU, a linear schedule, contrastive weight 0.05, temperature 0.1, and projection dimension 256. D2V-F0 uses one GPU, batch size four, and a cosine schedule. The locally trained W2V2-F1/F2 and D2V-F0 runs use seed 1337.

3.3. Calibration and evaluation

DADS computes one global native-coordinate ranking from labeled training speech, using every tenth utterance after sorting identifiers within each speaker. This gives 1,640/1,615/1,600 calibration utterances for Folds 0/1/2; these labels determine the ranking only, and development and test speech are never used for scoring.

Within each model-fold setting, all methods use the same reference, adapted model, CTC head, and retained-coordinate count. Final-layer features are cached in float32, and masked representations are decoded with greedy CTC without a language model. We report 25%, 50%, and 75% retention where available, use three fixed

Table 1. Main Fold-0 results. Test WER (%) is shown for wav2vec2 Large and data2vec Audio Large. Bold highlights the DADS results at each retained-shift budget.

Selection rule	Retained shift (%)	WER (%)
wav2vec2 Large		
Full	100	10.50
DADS	75	10.58
DADS	50	11.71
NoShift	0	118.91
data2vec Audio Large (Fold 0)		
Full	100	11.39
DADS	75	10.96
DADS	50	12.94
NoShift	0	13.77

mask seeds for Random, and compare reversion of the highest- and lowest-utility quarters. WER is total word edit distance divided by total reference words.

4. RESULTS

We organize the results around one question: are fine-tuning-induced representation changes uniformly useful to the adapted recognizer? We first establish that a compact, decision-aligned subset is sufficient, then compare selection rules at matched budgets. Counterfactual reversion tests necessity, while coordinate-frame and calibration analyses test the scope and practical stability of the effect.

4.1. DADS identifies recognition-critical adaptation changes

Fine-tuning does not make every representation coordinate equally important. The main Fold-0 comparison in Table 1 shows that the recognition benefit is concentrated in a subset of the induced shift: DADS can restore most of the adapted recognizer’s behavior without retaining the full change.

On the released wav2vec2 Large recognizer, retaining 75% of the shift with DADS recovers nearly the full adapted behavior: 10.58% WER versus 10.50% for Full. The paired difference is only 0.09 percentage points, with a 95% bootstrap interval of $[-0.21, 0.38]$. At the more compact 50% budget, DADS still preserves the adaptation benefit and substantially outperforms the magnitude heuristic (11.71% versus 16.64% WER).

The same pattern appears for data2vec Audio Large: DADS gives the lowest WER at both 75% and 50% retention. Across both backbones, the results indicate that adaptation information is non-uniformly distributed and that decision-aligned selection identifies a compact recognition-supporting subset.

4.2. Decision alignment outperforms heuristic selection

Retention alone cannot distinguish decision alignment from the generic benefit of keeping a fixed number of coordinates. Within each fold, we hold the model, head, test set, and retained budget fixed and compare the ranking rules directly at 50% in Table 2.

Decision alignment is the consistent source of the improvement: DADS is the best selection rule at the 50% budget on both held-out W2V2 folds. It improves over Magnitude by 0.62 and 0.91 percentage points on Folds 1 and 2, respectively, while also outperforming

Table 2. Test WER (%) at 50% retention for wav2vec2 Large on Folds 1 and 2. Random is the mean $\pm$ sample SD over three masks. Bold indicates the best WER in each row.

Fold	DADS	Magnitude	Gradient	Random
1	16.01	16.63	19.79	23.39 $\pm$ 0.58
2	15.58	16.48	20.48	24.20 $\pm$ 0.81

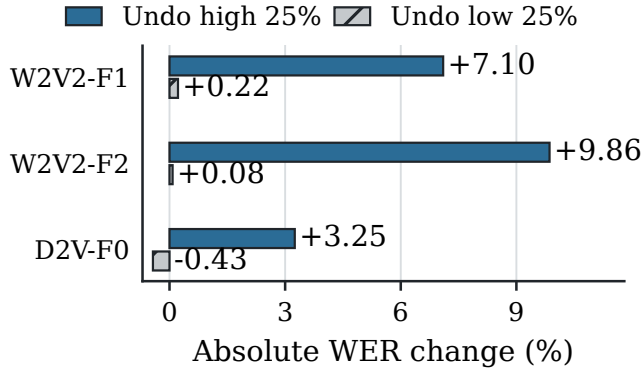


Fig. 2. Counterfactual WER change after undoing the highest- or lowest-scoring quarter of fine-tuning changes. Both interventions restore 256 native coordinates to the reference values and use the same adapted head. Positive values indicate worse recognition. Values denote absolute changes in WER, expressed in percentage units.

the gradient and random rankings. Because the coordinate count and frozen readout are identical, this comparison attributes the gain to the ranking criterion rather than to a larger representation or a retrained head.

4.3. DADS reveals causally important adaptation changes

The retention results establish sufficiency; a stronger test asks whether the selected changes are necessary for the adapted recognizer. We therefore revert groups of exactly the same size. As shown in Fig. 2, undoing the highest-utility quarter raises WER by 7.10/9.86/3.25 points for W2V2-F1/W2V2-F2/D2V-F0, whereas undoing the lowest-utility quarter changes WER by only 0.22/0.08/−0.43 points.

The interventions share the model, head, utterances, and coordinate count. Their contrast therefore reflects unequal functional reliance on native coordinates, rather than a generic tolerance to removing features. The training-derived ranking transfers to held-out speakers, supporting the view that DADS identifies functionally important adaptation changes.

4.4. DADS follows the native encoder–head coordinate geometry

The concentration is not a coordinate-free property of an arbitrary feature vector; it follows the native basis exposed by the trained recognizer. We test this by applying the equivalent parameterization $E' = QE$, $W'_a = W_a Q^\top$, with unchanged b_a , so that $W'_a E' + b_a = W_a E + b_a$. On W2V2-F0, five random orthogonal bases preserve the decoded predictions and WER (8.590% dev; 10.499% test), while the native ranking contains more Utility mass than the rotated null at

Table 3. W2V2-F0 Utility mass contained in the top native coordinates. Rotated values are mean $\pm$ SD over five equivalent orthogonal reparameterizations.

Top coordinates	Native	Orthogonal rotations
1%	7.78	3.37 $\pm$ 0.22
5%	20.23	12.68 $\pm$ 0.39
10%	30.30	21.55 $\pm$ 0.37
25%	50.93	42.38 $\pm$ 0.18

every Top- K level (Table 3). The top-1% Jaccard overlap is zero and the top-10% overlap is 0.058 ± 0.021 . DADS concentration therefore follows the coordinate organization established at the trained encoder–head interface.

4.5. DADS remains effective with limited calibration

The ranking is also stable when calibration data are limited. With only 128 utterances, DADS stays within 0.2 WER points of the full-calibration results at both 50% and 75% retention (13.11 $\pm$ 0.13% versus 12.94%, and 11.11 $\pm$ 0.10% versus 10.96%). Increasing the calibration set from 128 to 512 and 1,024 utterances raises the top-quarter overlap with the full ranking from 81.9% to 91.3% and 94.9%, respectively, without a meaningful change in WER. Thus, DADS does not require a large additional labeled calibration set to obtain a useful selection.

5. CONCLUSION

This paper proposes Decision-Aligned Delta Selection (DADS), a task-aware method for selecting the fine-tuning-induced representation shifts that matter to an adapted CTC recognizer. On L2-ARCTIC, DADS shows that adaptation gains are concentrated in a compact subset of the encoder’s native coordinates. For W2V2-F0, retaining 75% of the shift gives 10.58% WER, close to 10.50% with Full. At 50%, DADS achieves 11.71% WER, compared with 16.64% for magnitude selection, retaining 98.9% of the full adaptation gain. DADS also achieves the best 50% WER on both additional wav2vec2 folds and on data2vec Audio Large.

Reversion and rotation analyses further show that the selected shifts are functionally important and organized in the native encoder–head frame. Overall, DADS provides a simple way to identify useful adaptation changes without retraining the recognizer. Future work will test the method across more speakers, domains, backbones, and adaptation settings.

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